EAL TECH

SOC Home Lab Documentation

Version 2.0 | May 2026

Prepared by: Ezekiel A. Libara
Toronto, Ontario, Canada
EAL Tech IT Consulting

Section 1 — Lab Purpose and Goals

This document outlines the complete design, configuration, and build plan for the EAL Tech SOC Home Lab. The purpose of this lab is to simulate a real-world IT and cybersecurity environment for hands-on learning and practical skill development.

1.1 Lab Objectives

- Practice real SOC analyst work — detect, analyze and respond to incidents
- Build and manage an Active Directory environment with Windows Server 2022
- Learn how attackers think using a Red Team environment with Kali and Metasploit
- Practice CCNA networking using real managed switch and VLANs
- Deploy and configure Wazuh SIEM for real security monitoring
- Configure pfSense firewall for network perimeter security
- Build a professional portfolio hosted on GitHub

1.2 Lab Type

This is a Purple Team environment combining Blue Team defence and Red Team attack simulation.

- Blue Team — detection, monitoring, incident response, log analysis
- Red Team — attack simulation using Metasploit, Kali Linux, and C2 frameworks
- Purple Team — Red and Blue working together to improve defences

1.3 Documentation Storage

All lab documentation is stored in three locations for redundancy and professional visibility.

- Primary — Kingston NV3 500GB external drive (EAL-006)
- Backup — Google Drive cloud storage
- Portfolio — GitHub repository: EAL-Tech-Portfolio

Section 2 — Physical Asset Inventory

All physical devices are labeled with a unique Asset ID starting with EAL. Labels are physically attached to each device for easy identification.

Asset ID	Device	Model	RAM	Storage	Role	Location
EAL-001	HP Laptop	HP 15-fd0xxx	32GB	1TB	Blue Team Host	Bedroom Office
EAL-002	Intel NUC	NUC8i5BEH	16GB	500GB	Wazuh SIEM Host	Bedroom Office
EAL-003	ThinkPad T440p	20AN	16GB	500GB	Red Team Host	Bedroom Office
EAL-004	ThinkPad T470	20JN	8GB	240GB A400	Practice Machine	Bedroom Office
EAL-005	HP Aruba Switch	Managed	N/A	N/A	Network Core	Bedroom Office
EAL-006	Kingston NV3	500GB NVMe	N/A	500GB	External Storage	Bedroom Office
EAL-007	Powerline Adapter	TP-Link Kit	N/A	N/A	Network Bridge	Kitchen-Bedroom

Section 3 — Virtual Asset Inventory

All virtual machines are hosted on physical machines using VMware Workstation 16 Pro. Each VM has a defined role, operating system, RAM allocation, storage, and VLAN assignment.

VM ID	VM Name	Runs On	OS	RAM	Storage	Role	VLAN
VM-001	DC01	EAL-001	Windows Server 2022	4GB	60GB	Domain Controller	VLAN 10
VM-002	CLIENT01	EAL-001	Windows 11 Enterprise	2GB	40GB	Employee PC	VLAN 20
VM-003	CLIENT02	EAL-001	Windows 11 Enterprise	2GB	40GB	IT Helpdesk PC	VLAN 20
VM-004	CLIENT03	EAL-001	Windows 11 Enterprise	2GB	40GB	Manager PC	VLAN 20
VM-005	WAZUH01	EAL-002	Ubuntu 22.04	8GB	100GB	SIEM Server	VLAN 30
VM-006	KALI01	EAL-003	Kali Linux	4GB	60GB	Attacker Machine	VLAN 40
VM-007	SURICATA01	EAL-002	Ubuntu 22.04	4GB	40GB	IDS/IPS	VLAN 30
VM-008	OPENVAS01	EAL-002	Ubuntu 22.04	4GB	40GB	Vulnerability Scanner	VLAN 30
VM-009	PFSENSE01	EAL-001	pfSense	1GB	20GB	Firewall	Above all VLANs
VM-010	THEHIVE01	EAL-002	Ubuntu 22.04	4GB	50GB	Case Management	VLAN 30
VM-011	VELOCIRAPTOR	EAL-002	Ubuntu 22.04	2GB	40GB	Incident Response	VLAN 30

Section 4 — VLAN Network Plan

The HP Aruba managed switch is configured with 4 VLANs to isolate and segment network traffic. This mirrors real enterprise network design.

VLAN	Name	Purpose	Devices	Can Talk To
VLAN 10	Server	Active Directory, DNS, DHCP	DC01	VLAN 20 only
VLAN 20	Clients	User workstations	CLIENT01, CLIENT02, CLIENT03	VLAN 10 and VLAN 30
VLAN 30	Security	SOC monitoring tools	Wazuh, Suricata, OpenVAS, TheHive, Velociraptor	All VLANs — monitor only
VLAN 40	RedTeam	Attack simulation	Kali Linux with Metasploit	Isolated — controlled access only

Section 5 — IP Address Plan

All devices use static IP addresses. The third octet matches the VLAN number for easy identification.

Device	VLAN	IP Address	Subnet Mask	Gateway
DC01 — Active Directory	VLAN 10	192.168.10.10	255.255.255.0	192.168.10.1
CLIENT01	VLAN 20	192.168.20.10	255.255.255.0	192.168.20.1
CLIENT02	VLAN 20	192.168.20.11	255.255.255.0	192.168.20.1
CLIENT03	VLAN 20	192.168.20.12	255.255.255.0	192.168.20.1
WAZUH01	VLAN 30	192.168.30.10	255.255.255.0	192.168.30.1
SURICATA01	VLAN 30	192.168.30.11	255.255.255.0	192.168.30.1
OPENVAS01	VLAN 30	192.168.30.12	255.255.255.0	192.168.30.1
THEHIVE01	VLAN 30	192.168.30.13	255.255.255.0	192.168.30.1
VELOCIRAPTOR	VLAN 30	192.168.30.14	255.255.255.0	192.168.30.1
KALI01	VLAN 40	192.168.40.10	255.255.255.0	192.168.40.1
PFSENSE01 LAN	All VLANs	192.168.1.1	255.255.255.0	N/A

Section 6 — Complete Tool List

All tools are free and open source. Windows evaluation versions are free for 90-180 days.

6.1 Blue Team Tools

Tool	VM	Purpose	Cost
Windows Server 2022	VM-001	Active Directory, DNS, DHCP	Free 180 day trial
Windows 11 Enterprise	VM-002/003/004	Client workstations joined to domain	Free 90 day trial
Wazuh	VM-005	SIEM — log collection, alerting, dashboards	Free
Suricata	VM-007	IDS/IPS — network traffic detection	Free
OpenVAS	VM-008	Vulnerability scanning	Free
TheHive	VM-010	Incident case management	Free
Velociraptor	VM-011	Endpoint incident response	Free
Sysmon	All Windows VMs	Deep Windows event logging	Free
Wireshark	EAL-001	Packet capture and traffic analysis	Free
pfSense	VM-009	Firewall and network perimeter security	Free

6.2 Red Team Tools

Tool	VM	Purpose	Cost
Kali Linux	VM-006	Full attack platform with 600+ tools	Free
Metasploit Framework	VM-006 Kali	Exploit vulnerabilities in target systems	Free — built into Kali
Nmap	VM-006 Kali	Network scanning and port discovery	Free — built into Kali
Mimikatz	VM-006 Kali	Credential dumping and Windows attacks	Free
BloodHound	VM-006 Kali	Active Directory attack path mapping	Free
Netcat	VM-006 Kali	Network connections and shells	Free — built into Kali

Tool	VM	Purpose	Cost
Burp Suite	VM-006 Kali	Web application testing	Free community edition
SpiderFoot	VM-006 Kali	OSINT investigation and reconnaissance	Free

Section 7 — Physical Device Labels

Print and attach these labels to each physical device.

EAL-001 — HP Laptop

Field	Value
Asset ID	EAL-001
Device	HP Laptop 15-fd0xxx
Role	Blue Team Host
VMs Running	DC01, CLIENT01, CLIENT02, CLIENT03, pfSense
VLANs	VLAN 10 and VLAN 20
RAM	32GB
Storage	1TB

EAL-002 — Intel NUC

Field	Value
Asset ID	EAL-002
Device	Intel NUC8i5BEH
Role	SOC Monitoring Host
VMs Running	Wazuh, Suricata, OpenVAS, TheHive, Velociraptor
VLAN	VLAN 30
RAM	16GB
Storage	500GB

EAL-003 — ThinkPad T440p

Field	Value
Asset ID	EAL-003
Device	ThinkPad T440p
Role	Red Team Host
VMs Running	Kali Linux with Metasploit, BloodHound, SpiderFoot
VLAN	VLAN 40
RAM	16GB

Field	Value
Storage	500GB

EAL-004 — ThinkPad T470

Field	Value
Asset ID	EAL-004
Device	ThinkPad T470
Role	Practice Machine
Status	Needs Windows installation on A400 SSD
RAM	8GB
Storage	240GB Kingston A400

EAL-005 — HP Aruba Switch

Field	Value
Asset ID	EAL-005
Device	HP Aruba Managed Switch
Role	Network Core
VLANs Configured	VLAN 10, VLAN 20, VLAN 30, VLAN 40
Connected Devices	EAL-001, EAL-002, EAL-003, EAL-004

Section 8 — Lab Build Order

Follow this build order exactly. Do not skip phases. Tick the Status column when each task is complete.

Phase	Task	Device	Status
Phase 0	Label all physical devices	All	To Do
Phase 0	Buy TP-Link Powerline adapter kit	Kitchen to Bedroom	To Do
Phase 0	Connect all devices to switch with ethernet	EAL-005	To Do
Phase 1	Configure VLANs 10, 20, 30, 40 on switch	EAL-005	To Do
Phase 2	Download Windows Server 2022 ISO	EAL-001	To Do
Phase 2	Install Windows Server 2022 as VM-001 DC01	EAL-001	To Do
Phase 2	Configure Active Directory on DC01	EAL-001	To Do
Phase 2	Set static IP 192.168.10.10 on DC01	EAL-001	To Do
Phase 3	Download Windows 11 Enterprise ISO	EAL-001	To Do
Phase 3	Install 3 Windows 11 client VMs	EAL-001	To Do
Phase 3	Join all clients to ezekiel.lab domain	EAL-001	To Do
Phase 4	Install Wazuh on Ubuntu VM on NUC	EAL-002	To Do
Phase 4	Install Sysmon on all Windows VMs	EAL-001	To Do
Phase 4	Connect all Windows agents to Wazuh	EAL-002	To Do
Phase 5	Install Kali Linux on T440p	EAL-003	To Do
Phase 5	Configure Metasploit and attack tools	EAL-003	To Do
Phase 5	Install BloodHound for AD enumeration	EAL-003	To Do
Phase 6	Install Suricata IDS on NUC	EAL-002	To Do
Phase 6	Install OpenVAS vulnerability scanner	EAL-002	To Do

Phase	Task	Device	Status
Phase 6	Install TheHive case management	EAL-002	To Do
Phase 6	Install Velociraptor incident response	EAL-002	To Do
Phase 7	Install pfSense firewall VM on HP laptop	EAL-001	To Do
Phase 7	Configure pfSense VLAN routing rules	EAL-001	To Do
Phase 8	Run first Metasploit attack simulation	EAL-003	To Do
Phase 8	Investigate first alert in Wazuh	EAL-002	To Do
Phase 8	Create first case in TheHive	EAL-002	To Do
Phase 8	Write first incident report	EAL-001	To Do
Phase 9	Upload all documentation to GitHub	EAL-001	To Do

Section 9 — Lab Runbook

Write an entry every time you complete a task. This becomes your professional portfolio and shows employers real hands-on experience.

Runbook Entry Template

Field	Details
Date	
Task Completed	
Device / VM	
What I Did	
Problems Encountered	
How I Fixed It	
What I Learned	
Next Step	

First Runbook Entry — Example

Field	Details
Date	May 18, 2026
Task Completed	Created Complete Lab Documentation Version 2.0
Device / VM	EAL-006 Kingston NV3 External Drive
What I Did	Created complete lab documentation including all asset inventories, IP plan, VLAN plan, tool list and build order
Problems Encountered	None
How I Fixed It	N/A
What I Learned	How to plan a complete IT environment before building it — same way real companies do it
Next Step	Buy powerline adapter, label all physical devices, create GitHub account

Section 10 — Domain and Credentials

IMPORTANT: Keep this document secure. Do not share credentials publicly. Store only on EAL-006 external drive.

10.1 Active Directory Domain

Setting	Value
Domain Name	ezekiel.lab
Domain Controller	DC01
DC IP Address	192.168.10.10
DNS Server	192.168.10.10
DHCP Server	192.168.10.10
DHCP Scope VLAN 20	192.168.20.100 - 192.168.20.200

10.2 VM Credentials Template

VM	Username	Password	Notes
DC01	Administrator	Change me!	Domain admin — keep secure
CLIENT01	john.smith	Change me!	Regular employee account
CLIENT02	helpdesk	Change me!	IT helpdesk account
CLIENT03	manager	Change me!	Manager account
WAZUH01	admin	Change me!	Wazuh dashboard login
KALI01	kali	Change me!	Kali Linux default
PFSENSE01	admin	Change me!	pfSense web interface

Section 11 — GitHub Portfolio Plan

Your GitHub portfolio shows employers your real hands-on work. It is more powerful than any certificate.

11.1 Repository Structure

Folder	Contents	Purpose
Home-Lab-Documentation	This document, network diagram, IP plan	Shows planning and design skills
Threat-Intelligence	ShinyHunters investigation report	Shows research and analysis skills
Incident-Reports	All incident investigations from lab	Shows SOC analyst skills
CCNA-Notes	Network configurations and notes	Shows networking knowledge
Attack-Simulations	Metasploit lab reports	Shows red team understanding
Detection-Rules	Custom Wazuh and Suricata rules	Shows blue team skills

11.2 GitHub Repository URL

After creating your GitHub account — your portfolio will be at:

https://github.com/YourUsername/EAL-Tech-Portfolio

Section 12 — Network Connectivity

This section describes how all physical devices connect through the HP Aruba managed switch.

Device	Connection	Connected To	Switch Port
EAL-001 HP Laptop	Ethernet Cat5e/6	EAL-005 HP Aruba Switch	Port 1
EAL-002 Intel NUC	Ethernet Cat5e/6	EAL-005 HP Aruba Switch	Port 2
EAL-003 ThinkPad T440p	Ethernet Cat5e/6	EAL-005 HP Aruba Switch	Port 3
EAL-004 ThinkPad T470	Ethernet Cat5e/6	EAL-005 HP Aruba Switch	Port 4
EAL-007 Powerline Adapter 2	Ethernet Cat5e/6	EAL-005 HP Aruba Switch	Port 5
EAL-007 Powerline Adapter 1	Ethernet Cat5e/6	Home Router Kitchen	Router Port

Section 13 — Resources and Download Links

13.1 ISO Downloads

Resource	URL
Windows Server 2022	https://www.microsoft.com/evalcenter/evaluate-windows-server-2022
Windows 11 Enterprise	https://www.microsoft.com/evalcenter/evaluate-windows-11-enterprise
Kali Linux	https://www.kali.org/get-kali/
Ubuntu Server 22.04	https://ubuntu.com/download/server
pfSense	https://www.pfsense.org/download/

13.2 Tool Installation Links

Tool	URL
Wazuh	https://documentation.wazuh.com/current/installation-guide/
Suricata	https://suricata.io/download/
OpenVAS	https://greenbone.github.io/docs/
TheHive	https://docs.thehive-project.org/
Velociraptor	https://docs.velociraptor.app/
BloodHound	https://github.com/BloodHoundAD/BloodHound
Sysmon	https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon
Wireshark	https://www.wireshark.org/download.html

13.3 Learning Resources

Topic	Channel	Search Term
Active Directory	Josh Madakor	Josh Madakor Active Directory home lab
Wazuh SIEM	MyDFIR	MyDFIR Wazuh home lab
Metasploit	TCM Security	TCM Security Metasploit tutorial
SOC Analysis	MyDFIR	MyDFIR SOC analyst home lab
MITRE ATT&CK	AttackIQ Academy	AttackIQ MITRE ATT&CK beginner
CCNA	Jeremy's IT Lab	Jeremy's IT Lab CCNA free course
pfSense	Lawrence Systems	Lawrence Systems pfSense setup
BloodHound AD	TCM Security	TCM Security BloodHound tutorial

EAL Tech SOC Home Lab

Document Version 2.0 | May 2026

Ezekiel A. Libara | Toronto, Canada

EAL Tech IT Consulting